

Section 1: Data Warehouse

1. What is a Data Warehouse?
 - ☐ a) A system used for transactional processing
 - ☒ **b) A repository that stores historical and current data for analysis**
 - ☐ c) A collection of unrelated databases
 - ☐ d) A temporary storage area for operational data
2. Which of the following is NOT a characteristic of a Data Warehouse?
 - ☐ a) Subject-Oriented
 - ☐ b) Integrated
 - ☒ **c) Volatile**
 - ☐ d) Non-volatile
3. What is ETL in the context of Data Warehousing?
 - ☐ a) Extract, Transfer, and Load
 - ☒ **b) Extract, Transform, and Load**
 - ☐ c) Evaluate, Transfer, and Load
 - ☐ d) Extract, Transfer, and Log
4. A data warehouse follows which schema?
 - ☐ a) Relational schema
 - ☒ **b) Star schema, Snowflake schema, and Galaxy schema**
 - ☐ c) Hierarchical schema
 - ☐ d) Object-oriented schema
5. Which of the following is an example of a Data Warehouse?
 - ☐ a) MySQL
 - ☒ **b) Amazon Redshift**
 - ☐ c) MongoDB
 - ☐ d) SQLite
6. What is the purpose of an OLAP system?
 - ☐ a) Perform transactional processing
 - ☒ **b) Support analytical processing for decision-making**
 - ☐ c) Store data in an unstructured format
 - ☐ d) Replace OLTP systems
7. Fact tables in a data warehouse contain:

- a) Only primary keys
- b) Only foreign keys
- **c) Measures and foreign keys** ✓
- d) Only dimensions

8. What is Data Mart?

- **a) A small subset of a Data Warehouse designed for a specific department** ✓
- b) A database used in transactional processing
- c) A system that replaces a data warehouse
- d) A tool used for ETL processes

9. What does Granularity in Data Warehousing refer to?

- **a) Level of detail of data stored** ✓
- b) Data redundancy
- c) Size of the warehouse
- d) The number of tables in the database

10. Which type of OLAP stores precomputed data for fast query performance?

- a) ROLAP
- **b) MOLAP** ✗ (Your answer: c)
- c) HOLAP
- d) OLTP

11. Which tool is commonly used for Data Warehousing?

- a) Power BI
- **b) SAP BW** ✗ (Your answer: d)
- c) Tableau
- d) Excel

12. A Slowly Changing Dimension (SCD) is used to:

- a) Store real-time data
- **b) Manage changes in dimensional data over time** ✓
- c) Replace fact tables
- d) Improve data retrieval speed

13. The process of cleaning and transforming raw data before loading it into a data warehouse is called:

- **a) Data Cleansing** ✓

- b) Data Staging
- c) Data Mining
- d) Data Indexing

14. What is a Conformed Dimension?

- a) A dimension used only for a specific department
- **b) A dimension used across multiple fact tables and data marts** ✓
- c) A temporary table in a data warehouse
- d) A redundant dimension table

15. What is a surrogate key in a Data Warehouse?

- **a) A key that uniquely identifies each row** ✗ (Your answer: b)
- b) A primary key from a transactional database
- c) A key used for foreign relationships
- d) A key that duplicates the primary key

16. Which of the following is NOT a component of a Data Warehouse?

- a) Metadata
- b) ETL Process
- **c) Operational Database** ✓
- d) Fact Table

17. A "Star Schema" consists of:

- **a) A central fact table and multiple dimension tables** ✓
- b) Only normalized tables
- c) Only hierarchical relationships
- d) A large number of transactional tables

18. What is the primary role of Data Mining in a Data Warehouse?

- a) Data transformation
- **b) Discover patterns and insights from data** ✓
- c) Load data into the warehouse
- d) Cleanse redundant data

19. Which of the following is a benefit of Data Warehousing?

- a) Slower query processing
- **b) Centralized data for reporting and analysis** ✓
- c) Increased transaction processing speed
- d) Reduced data redundancy

20. The term "Drill Down" in Data Warehousing refers to:

- **a) Viewing detailed data from summary data** ✗ (Your answer: b)
- b) Summarizing detailed data into a higher-level view
- c) Removing duplicate records
- d) Data normalization

1. What is Informatica PowerCenter primarily used for?

- a) Transactional data processing
- **b) Extract, Transform, and Load (ETL) operations** ✓
- c) Database administration
- d) Web development

2. In Informatica, what is a Mapping?

- a) A connection between databases
- **b) A set of transformations to move data from source to target** ✓
- c) A backup process for data warehouse tables
- d) A tool used for writing SQL queries

3. What is the purpose of a Transformation in Informatica?

- a) Load data without any modification
- **b) Modify, filter, or aggregate data during ETL processing** ✓
- c) Delete records from the database
- d) Backup data before loading

4. What does a Source Qualifier Transformation do in Informatica?

- a) Inserts data into a target
- **b) Converts source data into a format compatible with Informatica** ✓
- c) Encrypts data for security
- d) None of the above

5. What type of transformation is Aggregator Transformation?

- a) Active Transformation ✓
- b) Active and Connected Transformation
- c) Passive Transformation
- d) Unconnected Transformation

6. In Informatica, what is a Workflow?

- a) A SQL query execution plan
- b) A process that runs ETL tasks in a sequence ✓
- c) A data modeling technique
- d) A type of relational schema

7. Which of the following is NOT a valid transformation in Informatica?

- a) Joiner Transformation
- b) Router Transformation
- c) Transaction Control Transformation ✓
- d) Lookup Transformation

8. What is a Lookup Transformation used for?

- a) To fetch related data from a different table or source ✓
- b) To perform aggregate functions
- c) To remove duplicates from source data
- d) To execute SQL queries

9. What is the purpose of Filter Transformation?

- a) To split data into multiple target tables
- b) To join two sources based on a condition
- c) To filter records based on a specified condition ✓
- d) To update target data

10. What type of transformation is Rank Transformation?

- a) Active Transformation ✗ (Your answer: b)
- b) Passive Transformation
- c) Semi-Active Transformation
- d) Dynamic Transformation

11. In Informatica, what does the Joiner Transformation do?

- a) Filters data from multiple sources

- **b) Joins data from two heterogeneous sources** ✓
- c) Creates a temporary table
- d) Splits data into different target tables

12. **What is Parameter File in Informatica?**

- a) A file used to store database queries
- **b) A file that stores values for workflow and session variables** ✓
- c) A file used to store temporary data
- d) A backup file for mappings

13. **What is the purpose of Stored Procedure Transformation?**

- a) To execute SQL queries
- **b) To call a stored procedure from the database** ✓
- c) To filter records based on a condition
- d) To execute PowerShell scripts

14. **What is a Session in Informatica?**

- **a) A connection between a source and target** ✗ (Your answer: a)
- **b) A task that loads and transforms data** ✓
- c) A method to write SQL queries
- d) A way to filter data before it reaches the target

15. **Which of the following is NOT an Active Transformation?**

- a) Filter Transformation
- b) Expression Transformation
- **c) Rank Transformation** ✗ (Correct answer: b)
- d) Aggregator Transformation

16. **What is the purpose of Sorter Transformation?**

- **a) To sort records in ascending or descending order** ✓
- b) To perform lookups on another table
- c) To filter data based on conditions
- d) To split data into multiple target tables

17. **What happens when a Persistent Cache is enabled in Lookup Transformation?**

- **a) The cache is reused across multiple sessions** ✓
- b) Data is refreshed automatically after each session
- c) The cache stores only new lookup values

- d) The lookup transformation runs without cache

18. In Informatica, Normalizer Transformation is used for:

- a) Aggregating and summarizing data
- **b) Converting columnar data into row-wise format** ✓
- c) Joining multiple data sources
- d) Sorting data before transformation

19. What is the function of Transaction Control Transformation?

- a) It is used to filter records
- b) It performs an SQL join
- **c) It defines commit and rollback transactions based on conditions** ✓
- d) It creates a new mapping

20. What is Recovery Strategy in Informatica?

- **a) A method to restart failed workflows from the point of failure** ✓
- b) A strategy to optimize session performance
- c) A method to roll back committed transactions
- d) A method to clean temporary data

1. What is Tableau primarily used for?

- A) Web Development
- B) Data Visualization** ✓
- C) Database Management
- D) Video Editing

2. Which of the following is NOT a file type in Tableau?

- A) .twbx
- B) .twb** ✗ (Correct answer: D) .xls)
- C) .tds
- D) .xls

3. What does a .twbx file contain?

- A) Only workbook metadata
- B) Workbook with data included** ✓
- C) Only dashboards
- D) Only SQL queries

4. Which of these is a measure in Tableau?

- A) Region
- B) Sales** ✓

- C) Product Name
- D) Country

5. What is the purpose of a calculated field in Tableau?

- A) To change data types
- B) To create custom calculations** ✓
- C) To clean raw data
- D) To filter data

6. What is the default aggregation for a measure in Tableau?

- A) SUM** ✓
- B) COUNT
- C) AVERAGE
- D) MIN

7. Which chart is best suited to show trends over time?

- A) Bar Chart
- B) Line Chart** ✓
- C) Pie Chart
- D) Scatter Plot

8. In Tableau, which type of filter applies to all worksheets using the same data source?

- A) Context Filter
- B) Extract Filter
- C) Data Source Filter** ✗ (Correct answer: **D) Global Filter**)

9. What is the function of the Pages shelf in Tableau?

- A) To break down data into pages for detailed analysis** ✓
- B) To create a hierarchy
- C) To format visualizations
- D) To filter data

10. What is a dual-axis chart in Tableau?

- A) A chart with two dimensions
- B) A chart with two measures on different axes** ✓
- C) A chart with two different dashboards
- D) A chart with two worksheets

11. What does the 'LOD Expression' stand for in Tableau?

- A) Level of Data
- B) Level of Detail** ✓
- C) Line of Data
- D) List of Dimensions

12. Which type of join returns only matching records from both tables?

- A) Inner Join ☒
- B) Left Join
- C) Right Join
- D) Full Outer Join

13. What is a blend in Tableau?

- A) A type of filter
- B) A method to combine multiple data sources ☒
- C) A dashboard feature
- D) A function for formatting

14. Which Tableau feature allows creating an interactive dashboard with clickable filters?

A) Actions ☒ (Your answer: B) (Correct answer: A) Actions))

15. What does an extract in Tableau do?

- A) Creates a static snapshot of data ☒
- B) Directly connects to a live database
- C) Merges different workbooks
- D) Creates a new calculated field

16. Which Tableau function is used to find the difference between two dates?

- A) DATEDIFF() ☒
- B) DATEADD()
- C) DATETRUNC()
- D) NOW()

17. What is the purpose of a hierarchy in Tableau?

- A) To define relationships between datasets
- B) To drill down from higher-level data to lower-level data ☒
- C) To change data formats
- D) To create multiple dashboards

18. Which of the following is NOT a type of filter in Tableau?

- A) Extract Filter
- B) Dimension Filter
- C) Worksheet Filter (Correct answer: C)
- D) Context Filter

19. What is the use of a parameter in Tableau?

- A) To create a dynamic value that can be changed ☒
- B) To filter data permanently
- C) To group similar values
- D) To change the workbook theme

20. Which of these databases can Tableau connect to?

- A) MySQL
- B) PostgreSQL
- C) SQL Server
- D) All of the above** ☒